

Juan David Muñoz-Bolaños

Application Engineer Image Processing

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Professional Summary

Dedicated R&D Optical Engineer with a robust background in combining **High-Precision Optical Metrology** with **Industrial Machine Vision** systems. Expert in developing advanced illumination concepts, real-time image processing algorithms (HALCON/Python), and hardware-software integration for automated inspection units. Proven problem-solving skills with a track record of bridging the gap between complex optical research and reliable industrial deployment.

TECHNICAL COMPETENCIES

Machine Vision Hardware

Mastering of CMOS/CCD camera integration, lens selection (telecentric, macro), and custom illumination design (Dark-field, Bright-field, coaxial).

Image Processing Libraries

Advanced proficiency in Halcon and Cognex VisionPro for real-time inspection, part recognition, and measurement. Familiar with OpenCV and Python-based vision pipelines.

System Alignment & Justage

Hands-on experience in the precise alignment of complex opto-mechanical setups (Laser sources, scanners, detectors). Expert in qualifying components from market-leading vendors.

Automation & Integration

Skilled in automating optical systems with Python, C++, and LabVIEW. Experienced in ensuring Gage R&R and measurement reliability in harsh industrial environments.

R&D Methodology

Experience bringing systems from concept to series prototypes, including ZEMAX modeling, procurement management, and technical documentation.

PROFESSIONAL EXPERIENCE

INTECOL S.A.S.

Junior Machine Vision System Engineer

Medellín, Colombia

Mar 2018 – Mar 2019

- **Vision Pipeline Engineering:** Designed and implemented Halcon-based beverage bottle inspection systems; developed custom illumination strategies to reliably detect liquid levels and printing defects.
- **Industrial Integration:** Commissioned vision systems using smart cameras and custom illumination for auto-part recognition at a car subdivision of Renault; integrated vision sensors with robotic arms for flame-surface treatment processes.
- **QA & Feasibility:** Performed systematic Gage R&R assessments and feasibility studies for new industrial clients, ensuring 24/7 reliability of vision components on the floor.

Medical University of Innsbruck

PhD Researcher – Optical Metrology & Automation

Innsbruck, Austria

Jan 2022 – Present

- **System Build & Justage:** Led the end-to-end assembly of a custom multiphoton microscope; adjusted galvo scanners, laser sources, and PMTs to achieved sub-micron resolution.
- **Adaptive Optics Control:** Programmed a Python/C++ control stack to synchronize scientific cameras with active wavefront correctors for real-time sample characterization.
- **Vendor & Component QM:** Managed technical specification sheets and qualified precision hardware from Thorlabs and Edmund Optics as part of system procurement.

- **Instrumentation Design:** Prototyping of a dispersed fiber-probe Raman spectrometer; handled optical layout justage and hardware-software calibration.
- **Data Evaluation Hooks:** Integrated computational data analysis (Python ML) with hardware sensors to optimize measurement throughput and validation.

EDUCATION

Medical University of Innsbruck <i>Ph.D. in Adaptive Optics & Microscopy</i>	Austria <i>Jan 2022 – Present</i>
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Friedrich Schiller University Jena <i>M.Sc. in Photonics</i>	Germany <i>2019 – 2021</i>
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University of Cauca <i>B.Sc. in Physics Engineering</i>	Colombia <i>2012 – 2018</i>
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PUBLICATIONS & AWARDS

- **Awards:** ASML & ZEISS Summer Schools participant; COLFUTURO Grant 2026 awardee.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte
Head of Institute of Biomedical Physics
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Dr. Christoph Krafft
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Leibniz Institute of Photonic Technology